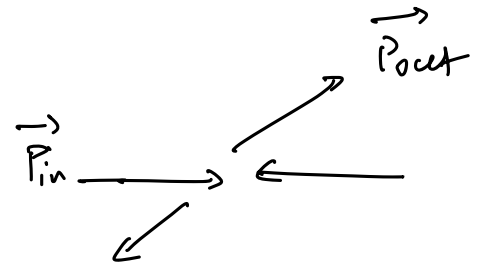


Cross section

center of mass

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{(E_1 + E_2)^2}$$

$$\frac{|\vec{P}_{out}|}{|\vec{P}_{in}|} |\mathcal{M}|^2$$



$$E_1 + E_2 = \sqrt{s}$$

$$\frac{d\sigma}{d\Omega} \propto \frac{1}{s}$$

$$\frac{dN}{dt} = \sigma \cdot \frac{dN_p}{dt} \cdot n_b \cdot d$$

$$\Rightarrow \Gamma(i \rightarrow f) = v_p \cdot \sigma$$

$$\sigma = \frac{1}{v_p} \Gamma(i \rightarrow f)$$

LAB frame w/ fixed target

$$\frac{d\sigma}{d\Omega} \propto \frac{1}{\sqrt{(\underline{P}_1 \cdot \underline{P}_2)^2 - (m_1 m_2)^2}} \int |\mathcal{M}|^2 \delta^{(4)}(\dots) \pi \frac{d^4 \underline{P}_i}{(2\pi)^4}$$

Lorentz invariant

$$\underline{P}_1 = (E_1, \vec{P}_1)$$

$$\vec{P}_{in} \equiv \vec{P}_1$$

$$\underline{P}_2 = (m_2, \vec{0})$$

$$\underline{P}_3 = (E_3, \vec{P}_3)$$

$$\underline{P}_4 = (E_4, \vec{P}_4)$$

$\vec{P}_{out} \equiv \vec{P}_3$
momentum of probe
after interaction

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{m_2^2} \frac{|\vec{P}_{out}|}{|\vec{P}_{in}|} |\mathcal{M}|^2$$

Rutherford experiment: $\alpha + N \rightarrow \alpha + N$

$$K_\alpha = 5 \text{ MeV} \ll M_\alpha = 3.7 \text{ GeV}$$

Compton scattering:

Möller scattering:

Bhabha scattering:

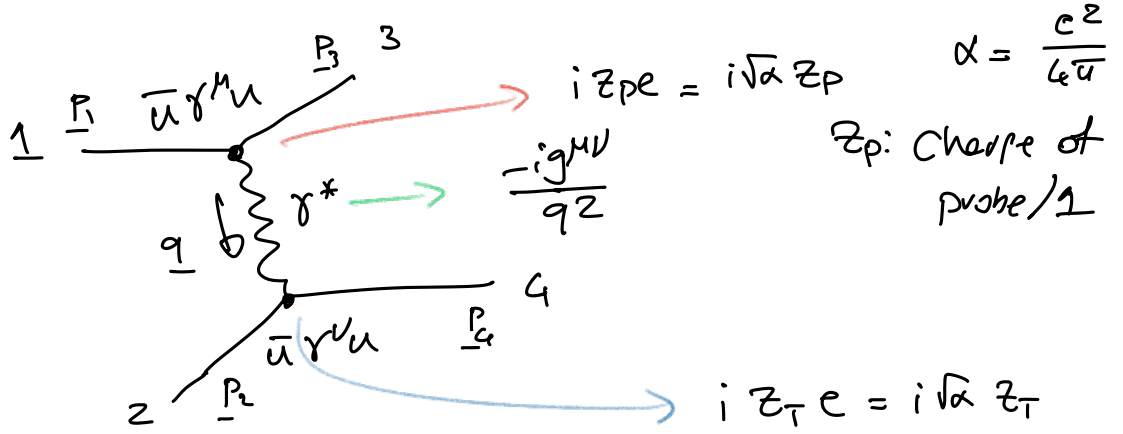
$$\gamma + e^- \rightarrow \gamma + e^-$$

$$e^- + e^- \rightarrow e^- + e^-$$

$$e^+ + e^- \rightarrow e^+ + e^-$$

} EM
interactions

$$|M|^2 = ?$$



$$\underline{p}_1 = \underline{q} + \underline{p}_3$$

$$\underline{q} = \underline{p}_1 - \underline{p}_3$$

$$|\underline{q}|^2 \neq 0 \Rightarrow \gamma^* \text{ virtual}$$

$$\Delta E \cdot \Delta t \sim \hbar \approx 1$$

uncertainty principle allows $|\underline{q}|^2 \neq 0$ for γ^*

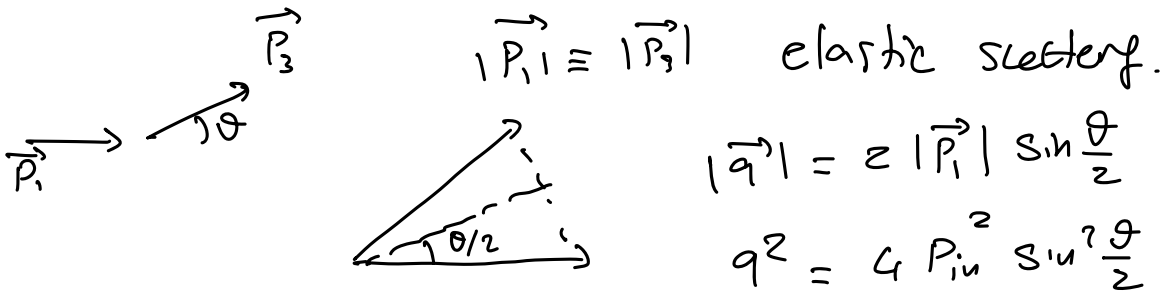
$$\mathcal{M} = \langle f | H_I | i \rangle$$

$$\mathcal{M} \propto \frac{\sqrt{\alpha} \sqrt{\alpha}}{q^2} \sim \frac{\alpha}{q^2}$$

Rutherford force: $\alpha + N \rightarrow \alpha + N$

$$K_\alpha = 5 \text{ MeV} \ll m_\alpha = 3.7 \text{ GeV} \ll \frac{A}{2} N \quad m_N \approx A \text{ GeV}$$

$$m_p \equiv m_1 \equiv m_\alpha \ll m_T \equiv m_2 \equiv m_N \approx 80 \text{ GeV}$$



$$|M|^2 = \left(\frac{\alpha m_T}{q^2} \right)^2 (m_1^2 + p_{in}^2 \cos^2 \frac{\theta}{2}) \quad m_T \equiv m_2, m_p \equiv m_1$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{m_1^2} \frac{|\vec{p}_3|}{|\vec{p}_1|} \frac{(\alpha m_1)^2}{q^4} (m_1^2 + p_{in}^2 \cos^2 \frac{\theta}{2})$$

elastic scattering.

highly non-relativistic
 $K_\alpha = 5 \text{ MeV}$

$$\frac{d\sigma}{d\Omega} \propto \frac{\alpha^2}{q^4} \sim \frac{\alpha^2}{16 p_{in}^4 \sin^4 \frac{\theta}{2}}$$

$$e^- + N \rightarrow e^- + N$$

$$K_e = 5 \text{ MeV}$$

$$m_e = 0.511 \text{ MeV}$$

$$E_e = 0.5 + 5 \text{ MeV} = 5.5 \text{ MeV}$$

$$\beta \approx 1 \quad \gamma \gg 1$$

highly relativistic

$$\begin{aligned} m^2 + p^2 \cos^2 \frac{\theta}{2} &= m^2 + p^2 (1 - \sin^2 \frac{\theta}{2}) = \\ &= \underbrace{m^2 + p^2} - p^2 \sin^2 \frac{\theta}{2} \\ &= E^2 - p^2 \sin^2 \frac{\theta}{2} = E^2 (1 - \frac{p^2}{E^2} \sin^2 \frac{\theta}{2}) \\ &= E^2 (1 - \beta^2 \sin^2 \frac{\theta}{2}) \end{aligned}$$

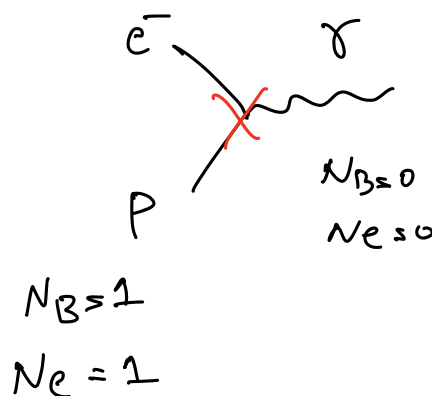
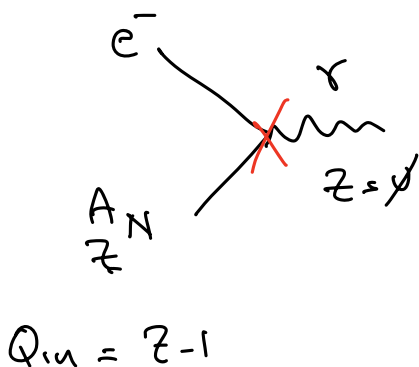
$$\frac{p}{E} = \beta$$

$$e^- + N \rightarrow e^- + N:$$

$$\frac{d\sigma}{d\Omega} = \underbrace{\left(- \right) \frac{1}{q^4} \frac{|\vec{p}_{out}|}{|\vec{p}_{in}|}}_{\text{Rutherford}} E_i^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

$$\frac{d\sigma}{d\Omega} \Big|_{\text{Mott}} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Ruth}} \times (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

Mott
cross
section

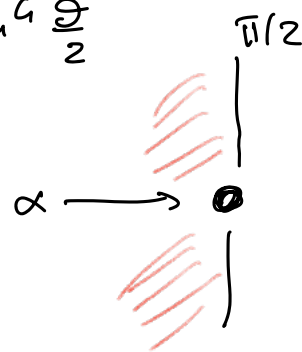
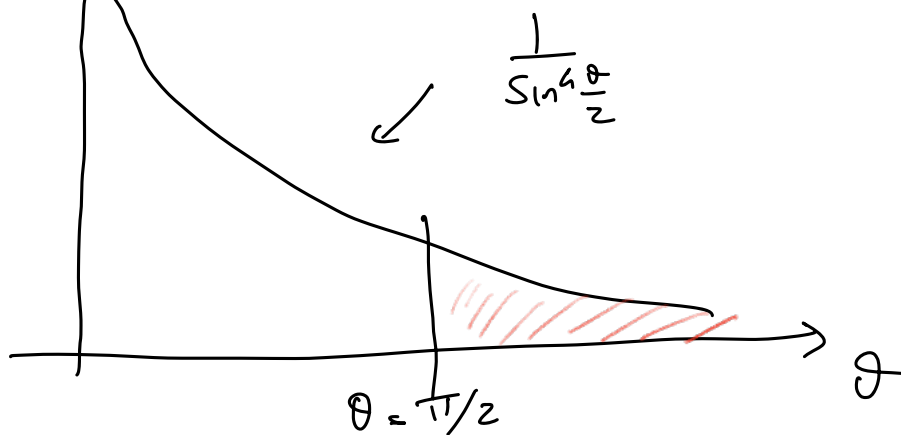


EM conserves
baryon numb.
lepton numb.

Rutherford formula

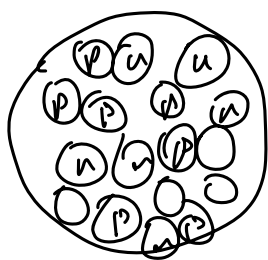
$$\frac{d\sigma}{d\Omega} \propto \frac{1}{\sin^4 \frac{\theta}{2}} = \frac{Z_p^2 Z_t^2 \alpha^2}{4 \sin^4 \frac{\theta}{2}}$$

events



$$\int_{\pi/2}^{\pi} \frac{d\sigma}{d\Omega} d\Omega = N(\theta > \pi/2) > 0 \Rightarrow \text{Back scattering}$$

$A \approx N$



nuclear force/Strong interaction
keeping nucleons within
nuclear radius.

$$R_N = r_0 A^{1/3} \quad r_0 \approx 1.1 \text{ fm}$$

Nucleons treated as Fermi gas:

gas confined within R_N (Z protons, $A-Z$ neutrons)

Fermi Energy $E_F = \frac{p_F^2}{2m_N}$

$p_F \approx 200 \text{ MeV}$. $E_F \approx 20 \text{ MeV}$
typical energy of nucleons
inside nucleus

to resolve nucleons (protons/neutrons)

$\Rightarrow$ probes with $20-30 \text{ MeV}$ energy.

Electrons with $p \approx 20 \text{ MeV}$

$$E = \sqrt{m^2 c^4 + p^2 c^2} \approx p \quad \beta \approx 1$$

$$\frac{d\sigma}{d\Omega} \propto \frac{\alpha^2}{p_{in}^4 \sin^4 \frac{\theta}{2}} E^2 (1 - \sin^2 \frac{\theta}{2}) \quad \beta = 1.$$

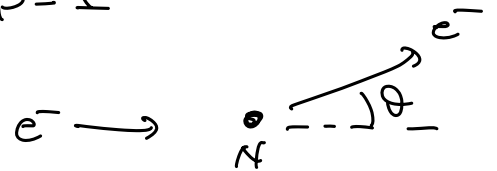
Assumptions:
pointlike probe
on
pointlike target

$H_I = EM$ Coulomb interaction.

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{measured}} \stackrel{?}{=} \left. \frac{d\sigma}{d\Omega} \right|_{\text{expected}}.$$

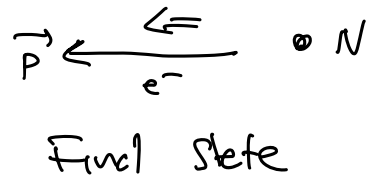
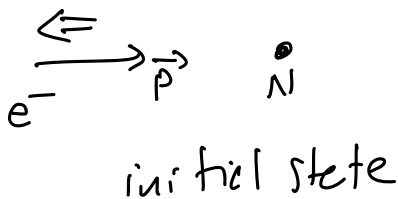
relativistic limit : $E_e \gg m_e \Rightarrow \beta = 1$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}} \propto \frac{\alpha^2}{q^4} E^2 (1 - \sin^2 \frac{\theta}{2}) \underbrace{\cos^2 \frac{\theta}{2}}$$



Max $\frac{d\sigma}{d\Omega}$: $\theta = \phi$

$$\theta = \pi \quad \frac{d\sigma}{d\Omega} \rightarrow \phi$$

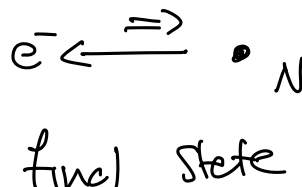
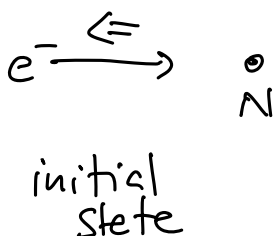


$$\text{helicity} \equiv h = \frac{\vec{p} \cdot \vec{S}}{|\vec{p}| \cdot |\vec{S}|}$$

massless particles: helicity invariant (same in all frames)

massive particles: helicity depends on the frame

helicity invariant for $\beta = 1$ electrons.



$\Delta J = 1$
not allowed
in EM

Mott cross section for pointlike particles.

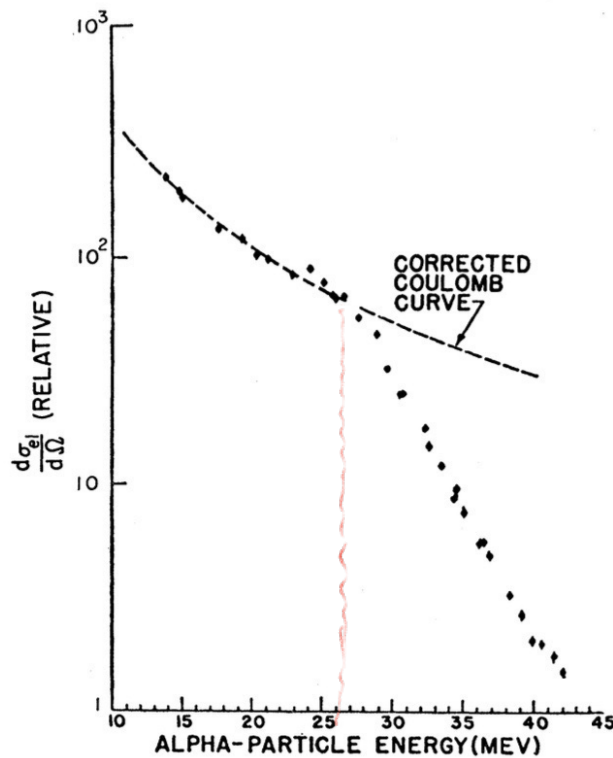
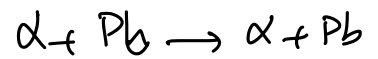


FIG. 5. Differential cross section for the elastic scattering of alpha particles by Pb at 60° as a function of the alpha-particle energy.



measured cross section
does not follow
Ruth. cross sec.

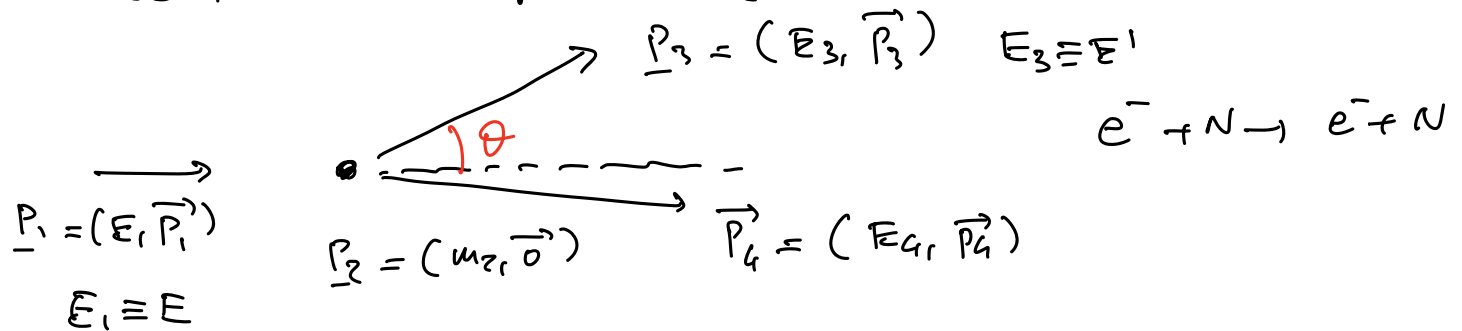
For $E \approx 20 \text{ MeV}$



Use electrons

with $E \geq 25 \text{ MeV}$

Mott and Rutherford formula do not include
recoil of the probe (assume $|\vec{P}_{in}| \approx |\vec{P}_{out}|$)



observables: E, E', θ (measured experimentally)

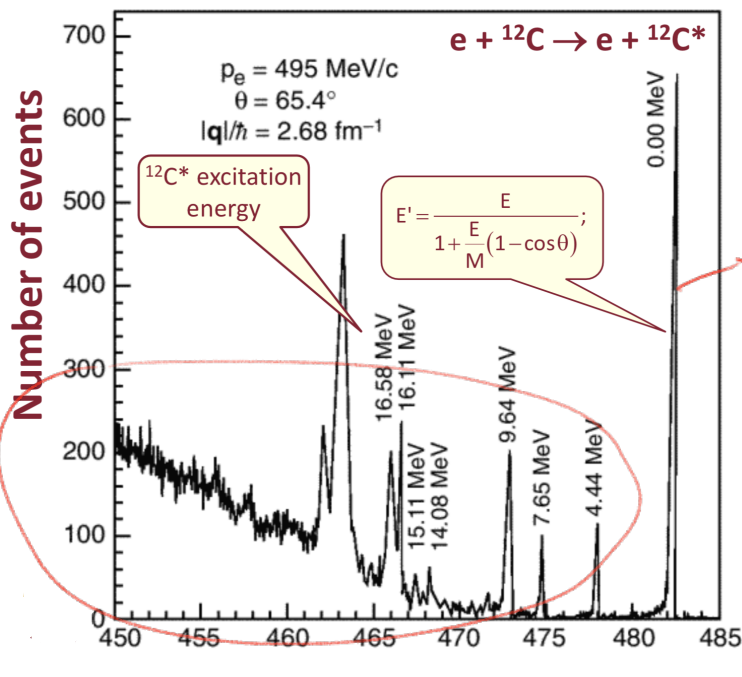
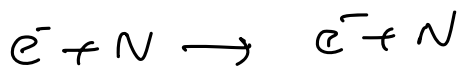
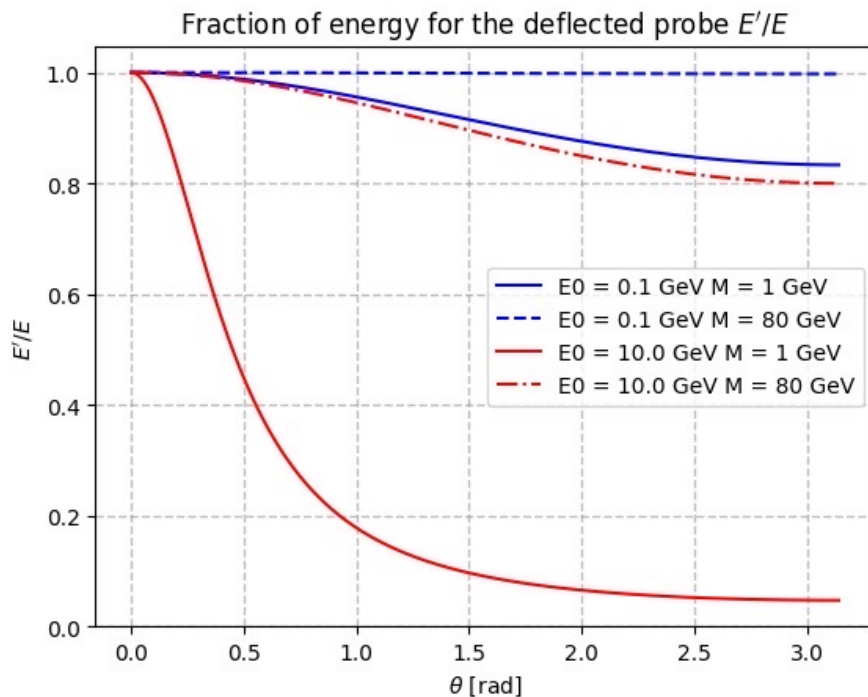
E : initial energy of e^-

E' : final en. of e^- (calorimeter or spectrometer)

θ : deflection angle

$$\underline{P}_1 + \underline{P}_2 = \underline{P}_3 + \underline{P}_4 \Rightarrow \underline{P}_4 = \underbrace{\underline{P}_1 - \underline{P}_3}_\uparrow + \underline{P}_2$$

$$E' = \frac{E}{1 + \frac{E}{M}(1 - \cos\theta)} = \frac{E}{1 + 2 \frac{E}{M} \sin^2 \frac{\theta}{2}}$$



Inelastic scattering

